PDS PRACTICAL 2

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Aim : Data Ingestion and Cleaning: Acquire data and prepare it for analysis.

Theory

1. What is data ingestion and why is it important? Which Python library is used?

Answer:

Data ingestion is the process of collecting, importing, and loading data from various sources into a system where it can be stored, processed, and analyzed. These sources can include structured data (CSV files, databases), semi-structured data (JSON, XML), and unstructured data (logs, images).

In data science, data ingestion is the first and foundational step of the data pipeline. If this step is poorly handled, every subsequent step—cleaning, analysis, and modeling—will be flawed.

Importance of Data Ingestion:

- Ensures data is available in a usable and structured format.
- Integrates data from multiple sources into one system.
- Maintains consistency and completeness of data.
- Enables real-time or batch data processing.
- Forms the base for accurate analysis and machine learning.

In Python, the most commonly used library for data ingestion and cleaning is Pandas. It provides powerful functions like:

- `read_csv()` for CSV files
- `read_excel()` for Excel files
- `read_json()` for JSON data

Example:

```
import pandas as pd
df = pd.read_csv("titanic.csv")
```

2. Why is data cleaning necessary?

Answer:

Data cleaning is the process of identifying and correcting errors, inconsistencies, and missing values in a dataset before performing analysis or applying machine learning algorithms.

Real-world data is rarely clean. It often contains:

- Missing values (e.g., unknown age in Titanic dataset)
- Duplicate records
- Incorrect or inconsistent formats
- Outliers or noise

If data is not cleaned:

- Analysis results become inaccurate or misleading.
- Machine learning models learn wrong patterns.
- Predictions become unreliable.

Benefits of Data Cleaning:

- Improves data quality and consistency.
- Increases accuracy of analysis.
- Enhances performance of machine learning models.
- Reduces bias caused by incorrect data.

Example (Titanic dataset): Missing values in the `Age` column must be handled:

3. How do you load a CSV file into a Pandas DataFrame?

Answer:

A CSV (Comma-Separated Values) file is one of the most common formats for storing tabular data. In Python, it can be loaded into a Pandas DataFrame using the `read_csv()` function.

This function reads the file and converts it into a structured format where:

- Rows represent observations.
- Columns represent features.

Basic Syntax:

```
import pandas as pd
df = pd.read_csv("titanic.csv")
```

Optional Parameters:

- `head()` → to preview data
- `index_col` → to set index column
- `na_values` → to define missing values

Example:

```
df = pd.read_csv("titanic.csv")
df.head()
```

4. What is the difference between `head()` and `tail()`?

Answer:

Both head() and tail() functions are used to quickly inspect the dataset, but they differ in which part of the data they display.

Function	Description
`head()`	Displays the first few rows (default = 5)
`tail()`	Displays the last few rows (default = 5)

Purpose:

- `head()` helps understand the structure and column names.
- `tail()` helps verify the end of the dataset and detect anomalies.

Example:

```
df.head()
df.tail()
```

You can also specify the number of rows:

```
df.head(10)
df.tail(3)
```

5. How can you identify missing values in a dataset?

Answer:

Missing values are common in real-world datasets and must be identified before cleaning. In Pandas, missing values are typically represented as `NaN` (Not a Number).

You can detect missing values using the following methods:

1. `isnull()` or `isna()`

Returns a Boolean DataFrame indicating missing values.

```
df.isnull()
```

2. Count missing values using `sum()`

```
df.isnull().sum()
```

This shows how many missing values are present in each column.

3. Check if any missing values exist

```
df.isnull().any()
```

4. Visual inspection

```
df.info()
```

This shows non-null counts for each column.

Example Insight (Titanic dataset):

- `Age` and `Cabin` columns often contain missing values.
- These must be handled before analysis.

1. What is data ingestion and why is it important in data science? Which Python library is commonly used for data ingestion and cleaning?

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

```
url = '/content/train (1).csv'
df = pd.read_csv(url)
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Tick
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A 211
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 175
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O 31012
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	1138
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	3734

```
df.isnull().sum()
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

```
age_column = df['Age']

age_column.head()
```

	Age
0	22.0
1	38.0
2	26.0
3	35.0
4	35.0

dtype: float64

```
df['date'] = '14 Apr 1912'

df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Tick
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A 211
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 175
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O 31012
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	1138
4	5	0	3	Allen, Mr. William Hen...	male	35.0	0	0	3734

```
sorted_df = df.sort_values(by='Fare', ascending=False)
sorted_df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticke
679	680	1	1	Cardeza, Mr. Thomas Drake Martinez	male	36.0	0	1	P 1775
258	259	1	1	Ward, Miss. Anna	female	35.0	0	0	P 1775
737	738	1	1	Lesurer, Mr. Gustave J	male	35.0	0	0	P 1775
88	89	1	1	Fortune, Miss. Mabel Helen	female	23.0	3	2	1995
438	439	0	1	Fortune, Mr. Mark	male	64.0	1	4	1995

```
sorted_df = df.sort_values(by='Fare', ascending=False)
sorted_df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticke
679	680	1	1	Cardeza, Mr. Thomas Drake Martinez	male	36.0	0	1	P 1775
258	259	1	1	Ward, Miss. Anna	female	35.0	0	0	P 1775
737	738	1	1	Lesurer, Mr. Gustave J	male	35.0	0	0	P 1775
88	89	1	1	Fortune, Miss. Mabel Helen	female	23.0	3	2	1995
438	439	0	1	Fortune, Mr. Mark	male	64.0	1	4	1995

```
df_renamed = df.rename(columns={
    'Sex': 'Gender',
    'Pclass': 'Passenger_Class'
})
df_renamed.head()
```

	PassengerId	Survived	Passenger_Class	Name	Gender	Age	SibSp	Pa
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	

```
df.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	89
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	3
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	4
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	1
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	3
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	51

```
filtered_df = df[(df['Age'] > 30) & (df['Fare'] > 50)]
filtered_df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticke
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	P 1759
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	11380
6	7	0	1	McCar thy, Mr. Timothy J	male	54.0	0	0	174
35	36	0	1	Holverson, Mr. Alexander Oskar	male	42.0	1	0	11378
52	53	1	1	Harper, Mrs. Henry Sleeper (Myna Haxtun)	female	49.0	1	0	P 1757


```
filtered_df = df[(df['Age'] > 30) & (df['Fare'] > 50)]
filtered_df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticke
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	P 1759
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	11380
6	7	0	1	McCarthy, Mr. Timothy J	male	54.0	0	0	174
35	36	0	1	Holverson, Mr. Alexander Oskar	male	42.0	1	0	11378
52	53	1	1	Harper, Mrs. Henry Sleeper (Myna Haxtun)	female	49.0	1	0	P 1757

```
grouped = df.groupby('Sex')['Fare'].mean()

grouped
```

	Fare
Sex	
female	44.479818
male	25.523893

dtype: float64

```
df2 = df[['PassengerId', 'Survived']]

merged_df = pd.merge(df, df2, on='PassengerId')

merged_df.head()
```

	PassengerId	Survived_x	Pclass	Name	Sex	Age	SibSp	Parch	Ti
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	2
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 1
				Heikkinen,					

```
df_dropped = df.drop(columns=['Cabin'])
df_dropped_rows = df.dropna(subset=['Age'])
df_dropped.head()
```

Conclusion:The successful completion of this practical on Data Ingestion and Cleaning demonstrates that the quality of data is the most critical factor in any data science pipeline. Through this exercise, we have reached the following conclusions:

Foundation of Analysis: Data ingestion is not merely about moving data; it is about establishing a reliable foundation. Without proper ingestion techniques using libraries like Pandas, subsequent analysis becomes impossible or inaccurate.

The Necessity of Cleaning: Raw data is almost always "noisy" or "dirty." We concluded that data cleaning is an essential step to handle missing values, remove duplicates, and correct inconsistencies. This process directly impacts the reliability of the insights derived.

Tool Proficiency: Using Python's Pandas library simplifies the transition from raw files (CSV, Excel, JSON) to structured DataFrames, allowing for efficient manipulation and preparation of data for machine learning or statistical modeling.